Thakur Trusts (Regd.)
THAKUR COLLEGE OF
ENGINEERING & TECHNOLOGY
Autonomous College Affiliated to University of Mumbai
Approved by All India Council for Technical Education (AICTE) and Government of Maharashita(GOM)
Conferred Autonomous Status by University Grants Commission (UGC) for 10 years w.e.f. A.Y. 2019-20
(among the First in the country to be conferred Autonomous Status) and awarded 'A+' Grade by NAAC (3rd Cycle)
(Institute Ranked 58th in India by NIRF, Ministry of Education, Government of India)
Estd. 2001

IN-SEMESTER EXAMINATION-I

F.E./F.T. (Semester-I)

SUBJECT – Mathematics-I

Branch: COMP, CIVIL, E&CS, CSE, AI&DS, IoT **Date: 24/10/2024**

Div.: ALL **Timing: 2:00 PM to 03:00 PM**

Duration: 60 Minutes **Maximum Marks: 20**

Instructions –

1. All questions are compulsory.
2. Assume suitable data wherever necessary and state the assumptions made.
3. Diagrams / sketches should be given wherever necessary.
4. Use of logarithmic table, drawing instruments and non-programmable calculators is permitted.
5. Figures to the right indicate full marks.

Q.1		Answer any 5 of the following questions	Marks	Course Outcomes	Learning Levels
	a.	If $f(x) = \cos x$ in $[-\pi, \pi]$ then find c of R.M.V.T.	2	1	R
	b.	Evaluate complex conjugate of $\frac{5+i}{2-i}$	2	3	R
	c.	Determine whether the series is convergent or divergent $\sum_{n=1}^{\infty} \frac{n!}{2^n}$	2	1	R
	d.	If $u = y^2e^x + x^2y^3$ then find value of $\frac{\partial^2 u}{\partial y \partial x}$.	2	2	R
	e.	Write polar form of $z = \frac{1}{2} + i\frac{\sqrt{3}}{2}$	2	3	R
	f.	If $\phi = 2x^2y - 4xy^4$ then find $\nabla\phi$ at (0, 1, 2).	2	2	U
	g.	Expand $f(x) = e^x$ in powers of $(x-2)$ using Taylor's series.	2	1	R
Q.2	a.	Find the directional derivative of $\varphi = 4xz^2 - 3x^2y^2z$ at (2, -1, 2) in the direction of $2i + 3j + 6k$.	5	2	A
		OR			
	b.	If $f\left(\frac{y-x}{yx}, \frac{z-x}{zx}\right)$, then show that $x^2\frac{\partial u}{\partial x} + y^2\frac{\partial u}{\partial y} + z^2\frac{\partial u}{\partial z} = 0$	5	2	A
Q.3	a.	Prove that $\frac{1}{1-\frac{1}{1-\frac{1}{1+\sinh^2 x}}} = -\sinh^2 x$	5	3	A
		OR			
	b.	Illustrate $\lim_{x \to \frac{\pi}{2}} (\sin x)^{\tan x}$	5	1	U